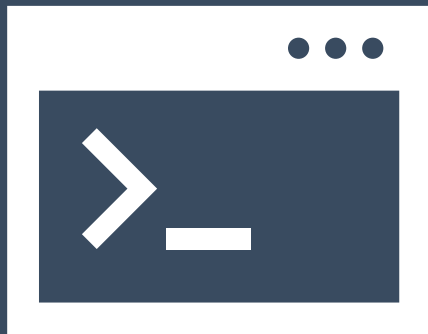


Introduction to Single-cell RNA-seq analysis

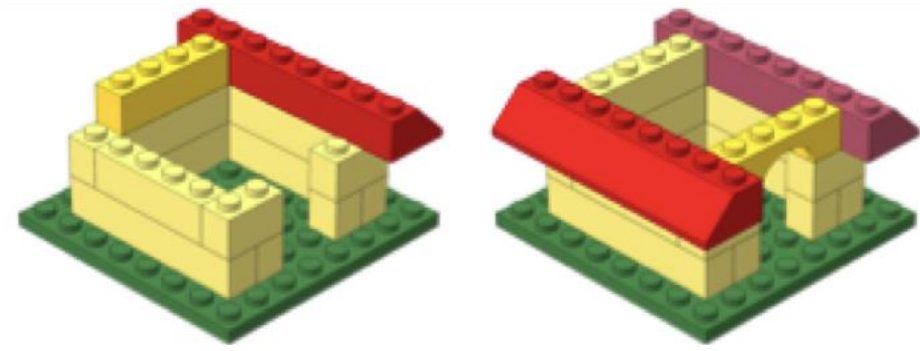
<https://tinyurl.com/hbc-scrnaseq-online>



Harvard Chan Bioinformatics Core



Workshop Scope



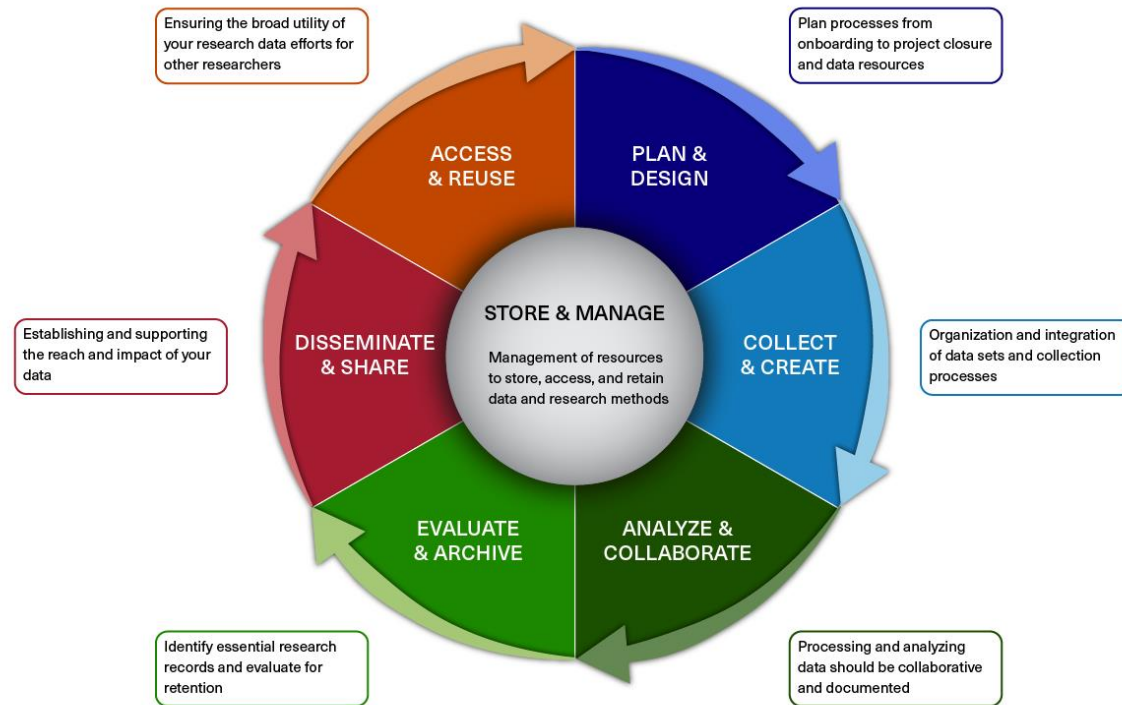
- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Exit survey

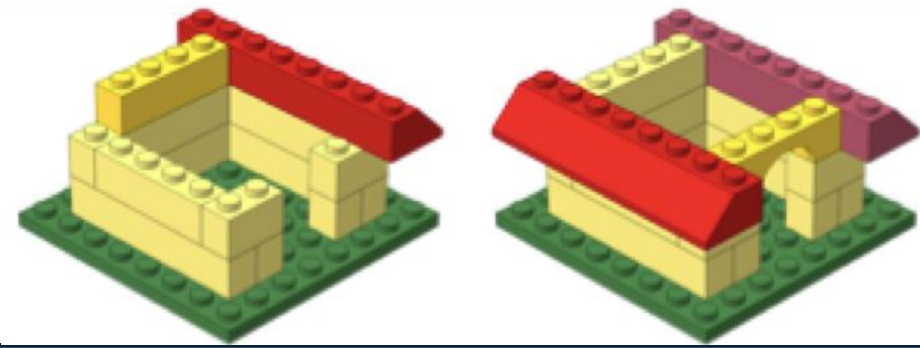
<https://tinyurl.com/scRNAseq-online>

Research Data Management (RDM)

BIOMEDICAL RESEARCH DATA LIFECYCLE



Useful resources



❖ Computational packages for single-cell analysis:

- ❖ <http://bioconductor.org/packages/devel/workflows/html/simpleSingleCell.html>
- ❖ <https://satijalab.org/Seurat/>
- ❖ <https://scanpy.readthedocs.io/>
- ❖ <https://github.com/seandavi/awesome-single-cell>

❖ Online courses:

- ❖ <https://hemberg-lab.github.io/scRNA.seq.course/>
- ❖ <https://github.com/SingleCellTranscriptomics>

❖ Resources for scRNA-seq Sample Prep:

- ❖ <https://www.protocols.io/>
- ❖ <https://support.10xgenomics.com/single-cell-gene-expression/sample-prep>
- ❖ <https://community.10xgenomics.com/>

Better RDM practice benefits you

- ❖ **HMS Data Management LMA**

- ❖ **Webpage:** <https://datamanagement.hms.harvard.edu>

- ❖ **Sign up for quarterly email updates**

- ❖ **Harvard-wide Research data Management**

- ❖ <https://researchdatamanagement.harvard.edu/>

Fall 2025 Data Lifecycle Training

Plan & Design

September 11 
Intro to O2

September 17 
Foundations in Shell

September 30 
Managing Research
Data Efficiently

October 16 
Project and Lab
Onboarding

November 25 
Writing a Data Management
and Sharing Plan

Collect & Analyze

September 18 
Intro to MATLAB

October 2 
O2 Portal

October 15 
Finding and Summarizing
Data from Colossal Files

November 5 
Intro to GIS, Maps & Data

November 6 
Intro to Python

November 19 
Tips and Tricks for the
O2 Cluster

Store & Evaluate

October 7 
Roadmap for Data Retention
Policies & Practices

October 16 
Introduction to the
General Records Schedule

October 29 
Data Horror Stories:
Avoid a Research Nightmare

November 13 
Off-Site Records Storage

December 2 
Shared Drives and Email

Share & Publish

December 3 
Data Sharing in
Repositories

December 9 
Data Organization and
Sharing in Open Science
Framework



In-person

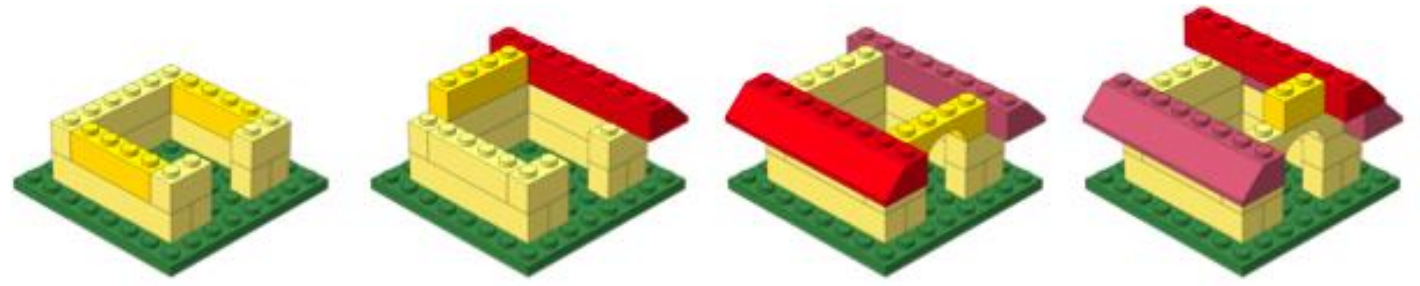


Virtual



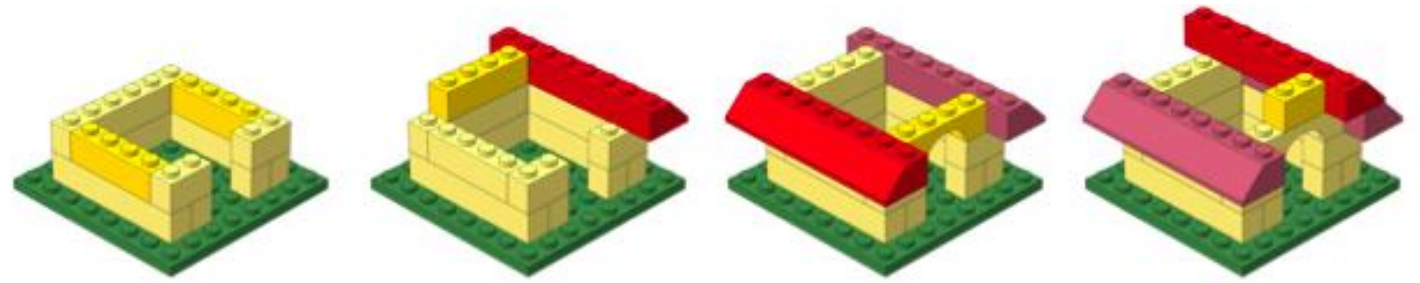
LONGWOOD
RESEARCH DATA MANAGEMENT

Keep building!



Module	Pre-requisite	Date	Time
Foundations in Shell	None	9/17/25	1-4pm
Needle in a Haystack	Foundations in Shell	10/15/25	1-4pm
Tips and Tricks for the O2 Cluster	Foundations in Shell	11/19/25	1-4pm

Keep building!



Workshop	Pre-requisite*	Dates	Time	Location
Pseudobulk and related approaches for scRNA-seq analysis	Introduction to R	October 21, 24, 28, 31	10am-12pm	Zoom
Introduction to Differential Gene Expression Analysis	Introduction to R	November 14, 18, 21, 25	10am-12pm	Zoom

<https://hsph.harvard.edu/research/bioinformatics/training/training-schedule/>

Talk to us early!

Involvement in study design to optimize experiments



Join us for HBC Community Breakfast!

- ❖ An opportunity to get to know others in the community
- ❖ Free food and beverages
- ❖ Great conversations
- ❖ Thursday, October 30th, 2025
 - ❖ 9:30 to 11am
- ❖ Raffle for a free workshop



More Information

- ❖ *HBC training materials: <https://hbctraining.github.io/main>*
- ❖ *HBC website: <http://bioinformatics.sph.harvard.edu>*

Contact Us

Sign up for our mailing list:

<https://tinyurl.com/hbc-training-mailing-list>

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu